SCTP AI ENGINEERING · CAPSTONE · TEAM 1

Kaya Kopi ops copilot: who does what, in what order

Objective: an assistant that answers ops questions from SQL, outlet spreadsheets, scanned invoices and chats, with the source shown for every answer. Read each row left to right by week. Follow the red CP markers for the critical path.

DEMO DAY
Sat 21 Nov 2026
5-min pitch · 5-min live demo · panel Q&A
Code freeze Fri 20 Nov

A4 step ID (letter = owner) · needs A3 must be done first · → C hands its output to that role · ☑ graded or checkpoint deliverable · CP1 critical path, in order (red outline)

	WEEK 1 · to Thu 5 Nov Checkpoint: data audit · ER diagram · trap list · architecture proposal · all 4 have committed	WEEK 2 · to Sat 14 Nov Checkpoint: ingest + retrieve on 2 sources (one is images) · 1 cited answer · OCR pipeline · spend check	WEEK 3 · to Fri 20 Nov Freeze: agent layer · guardrails + 20 evals passing · data quality report · hostile test · pitch timed	FINAL DELIVERABLES
A **Data Lead** POS SQL, outlet files, roster, clean schema Owner: [name]	**A1** Audit every data file — start day 1 ☑ data zip → rows, columns, dates, outlets **A2** Import POS dump to raw schema — needs A1 kayakopi_pos.sql → raw tables, counts match **A3** Hunt traps, with evidence — needs A2 ☑ CP1 raw + outlet files → trap list (query + result) **A4** Publish clean schema + ER — needs A3 → B C D ☑ CP2 trap list → schema.sql, ER diagram, column contract	**A5** Map outlet IDs, load outlet_daily — needs A4 8 outlet files → outlet_daily, AMK-03 restored **A6** Remove PII, restricted DB user — needs A4 → C ☑ roster → staff_roster_safe, read-only app user **A7** Build clean views, reconcile — needs A5, B5 → C CP3 raw + A5 + B5 → views, raw total = clean total **A8** Write 10 golden queries — needs A7 → D clean views → golden.sql with expected values	**A9** Finish data quality half, fix bugs — needs A3, A7 → D ☑ eval failures → report section, patches	· docs/data_audit.md and trap list · ER diagram and clean schema · Clean tables, AMK-03 restored · PII-safe roster, restricted DB user · Half of the data quality report
B **Evidence Lead** Invoices, docs, supplier price truth Owner: [name]	**B1** Audit scans and chat/email docs — start day 1 ☑ 70 scans + docs → audit sheet, traps found **B2** Design extraction prompt + schema — needs B1, D1 5 sample scans → tested prompt, cost per scan **B7** Build searchable docs index — needs B1 WhatsApp + emails → full-text index, search_docs()	**B3** Batch-extract all 70 invoices — needs B2 B2 prompt → cached JSON per invoice, failure log **B4** Validate all 70, by hand + sums — needs B3 ☑ JSON + scans → validated invoice_lines.csv **B5** Load lines, build price-truth view — needs B4, A4 → A C CP3 B4 + A4 contract → v_supplier_price_truth **B6** Reconcile purchases vs invoices (GST) — needs A2, B5 → D ☑ purchase table + B5 → v_purchase_reconciliation **B8** Hand over image + doc tools — needs B5, B7 → C B5 + B7 → get_invoice_image(), search_docs()	**B9** Finish data quality half, fix extraction — needs B6 → D ☑ eval failures → report section, patches	· 70 invoices extracted and validated · Supplier price-truth view · GST reconciliation, gap explained · Docs search tool, invoice image tool · Half of the data quality report
C **Agent & App Lead** Agent loop, tools, UI, state Owner: [name]	**C1** Repo skeleton + tool contract — start day 1 → D README, .env.example → runnable repo, return format **C2** Agent loop on stub tables — needs C1 C1 → loop answers one stub question **C3** Draft query sequence diagram — needs C2 → D ☑ C2 → Mermaid sequence, component notes	**C4** Build data_coverage + guarded run_sql — needs A4, A6 A4 contract, A6 user → 2 tested tools **C5** Build metric tools — needs A7, B5 CP4 A7 + B5 views → margin, price change, wastage **C6** Citations + no-data reply in code — needs C4, C5 C4, C5 → sources appended, refuses to guess **C7** First cited end-to-end answer — needs C5, C6 ☑ CP5 C5, C6 → one answer showing its source	**C8** Streamlit UI + evidence panel — needs C6, B8 C6, B8 → chat with SQL, trace, invoice image **C9** Session state + restart test — needs C2 C2 → SQLite sessions, follow-ups resolve **C10** Integrate guardrails + logging — needs D5, D6, B8 CP6 D5, D6, B8 → full pipeline, all tools wired **C11** Fix failures, freeze Fri 20 Nov — needs C10 ☑ CP7 D8, D9 → stable demo build	· Agent loop with all tools · Cited answers, honest "can't tell" · Evidence UI with invoice image · Restart-safe sessions · Frozen demo build
D **Quality & Delivery Lead** Guardrails, evals, logs, docs, pitch Owner: [name]	**D1** Set up process + budget tracker — start day 1 README, checklists → rules, spend sheet, team size confirmed **D2** Write guardrail spec — needs A3 A3 PII findings → rules, kill-shot in 5 phrasings **D3** Assemble architecture proposal — needs A4, C3 ☑ A4, C3 → architecture.md with 2+ Mermaid diagrams **D4** Build eval harness — needs C1 C1 contract → one-command runner, 3 sample cases	**D5** Code input + output guardrails — needs D2 → C D2 → guardrails.py with unit tests **D6** Add error wrapper, logs, spend — needs C2 → C C2 → JSONL traces, spend script **D7** Write eval cases (10 → 20+) — needs A8, B6 ☑ A8, B6 → cases.yaml, 2+ refusals, trap cases **D8** Run evals, route failures — needs C7, D7 C7 on → pass-rate trend, owner per failure	**D9** Hostile test, kill one dependency — needs C10 ☑ C10 → break list, screenshots of honest errors **D10** Report, post-mortem, pitch — needs A9, B9, C11 ☑ A9, B9, C11 → 1-page report, post-mortem, timed deck	· Guardrails in code, tested · 20+ evals, one command · Logs and API spend tracker · Architecture doc, 2+ Mermaid diagrams · Pitch, post-mortem, final report

AGREE ON DAY ONE (31 OCT)

1. Schema contract. A and B publish table and column names within 48 hours. C and D build on stub tables until then.

2. Tool return format. Every tool returns {data, sources[], caveats[]}. C owns it, D writes eval checks against it.

3. Repo layout. etl/ agent/ app/ evals/ docs/. data/ and .env stay git-ignored.

CRITICAL PATH

CP1 A3 → CP2 A4 → CP3 A7 + B5 → CP4 C5 → CP5 C7 → CP6 C10 → CP7 C11

If A4 slips, C and D are blocked. Publish a draft contract early and build on stubs.

If B3 or B4 slips, every margin answer slips. Test the extraction prompt on 5 scans in week 1.

WORKING RULES

Branches. One per person. Every pull request is reviewed by another role (A with C, B with D).

Commits. Real work from all 4. Pair work carries a Co-authored-by line.

Budget. D posts API spend weekly. No bulk vision run before the 5-scan test.

Evals. A failing case becomes a fix the same week.